41. Draw the structure of each of the following compounds: **(a)** ββ-butyrolactone; **(b)** ββ-valerolactone; **(c)** δδ-valerolactone; **(d)** ββ-propiolactam; **(e)** αα-methyl-δδ-valerolactam; **(f)** *N*-methyl-γγ-butyrolactam.

43. Formulate a mechanism for the reaction of methyl 9-octadecenoate with 1-dodecanamine shown on p. 984.

46. For each of the naturally occurring lactones below, draw the structure of the compound that would result from hydrolysis using aqueous base.

Sedanenolide, major contributor to the flavor of

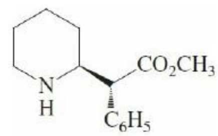
47.

a. *N*,*N*-Diethyl-3-methylbenzamide (*N*,*N*-diethyl-*m*-toluamide), marketed as DEET, is perhaps the most widely used insect repellent in the world and is especially effective in interrupting the transmission of diseases by mosquitoes and ticks. Propose one or more preparations of DEET (see structure below) from 3-methylbenzoic acid and any other appropriate reagents.

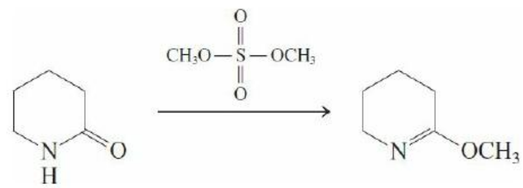
b. Icaridin (also called picaridin, see structure below) is another repellant useful against biting bugs. Like DEET, icaridin disrupts mosquitoes' odorant receptors, rendering them incapable of detecting humans. What is the name of the carbonyl-containing functional group in icaridin? What would you expect of the reactivity of its carbonyl group compared to that of the other carboxylic acid derivatives discussed in this chapter? Explain your answer.

48.

a. Like many chiral pharmaceuticals, Ritalin, widely used to treat attention deficit disorder, is synthesized and marketed as a racemic mixture. A synthetic route to the active enantiomer (see structure below) begins with the reaction of dimethyl sulfate with a six-membered ring lactam. Propose a mechanism for this reaction.



Ritalin
(Active stereoisomer)



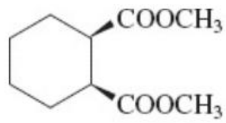
b. How many stereocenters does Ritalin contain? What are their stereochemical designations (*R* or *S*)? How many stereoisomers does Ritalin have, and how are they structurally related?

49. Formulate a mechanism for the formation of acetamide, $CH_3\overset{\overset{O}{\|}}{C}NH_2$, from methyl acetate and ammonia.

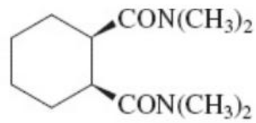
50. Give the products of the reactions of pentanamide with the reagents given in Problem 39, parts (a, e, f). Repeat for *N*,*N*-dimethylpentanamide.

51. Formulate a mechanism for the acid-catalyzed hydrolysis of 3-methylpentanamide shown on p. 991. **(Hint:** Use as a model the mechanism for general acid-catalyzed addition–elimination presented in Section 19-7.)

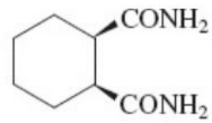
54. Upon treatment with LDA followed by protonation, compounds A and B undergo cis-trans isomerization, but compound C does not. Explain.



A



B



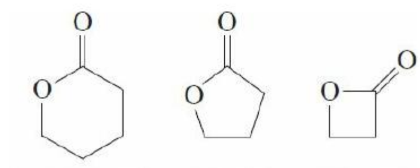
C

55. 2-Aminobenzoic (anthranilic) acid is prepared from 1,2-benzenedicarboxylic anhydride (phthalic anhydride) by using the two reactions shown here. Explain these processes mechanistically.

56. On the basis of the reactions presented in this chapter, write reaction summary charts for esters and amides similar to the chart for acyl halides (Figure 20-1). Compare the number of reactions for each of the compound classes. Is this information consistent with your understanding of the relative reactivity of each of the functional groups?

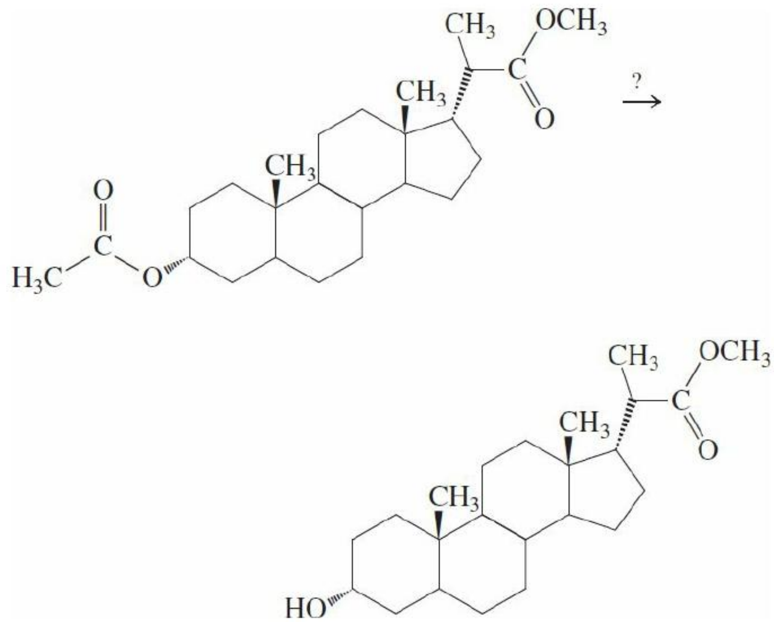
57. Show how you might synthesize chlorpheniramine (see also p. 762), a powerful antihistamine used in several decongestants, from each of carboxylic acids A and B. Use a different carboxylic amide in each synthesis.

58. Although esters typically have carbonyl stretching frequencies at about v˜=1740cm-1$\tilde{v} = 1740\ \text{cm}^{-1}$ in the infrared spectrum, the corresponding value for lactones can vary greatly with ring size. Three examples are shown below. Propose an explanation for the increasing energies of the carbonyl stretching vibrations with decreasing ring size of the lactone.



$\tilde{v} = 1735\ \text{cm}^{-1}$ 1770 cm^{-1} 1840 cm^{-1}

60. **CHALLENGE** Show how you would carry out the following transformation in which the ester function at the lower left of the molecule is converted into a hydroxy group but that at the upper right is preserved. **(Hint:** Do not try ester hydrolysis. Look carefully at how the ester groups are linked to the steroid and consider an approach based on transesterification.)



67. Friedel-Crafts acylations are best carried out with acyl halides, but other carboxylic acid derivatives undergo this process, too, such as carboxylic anhydrides or esters. These reagents may have some drawbacks, however, the subject of this problem.

Before you start, discuss as a group the mechanisms for forming acylium ions from acyl halides and carboxylic anhydrides in Section 15-13. Then divide your group in two and analyze the outcome of the following two reactions. Use the NMR spectral data given to confirm your product assignments. **(Hint:** D is formed via C.)